

JABALPUR ENGINEERING COLLEGE
JABALPUR, MP



REPORT ON
Parallelogram Linkage:
PANTOGRAPH

BACHELORS OF TECHNOLOGY
IN
MECHATRONICS ENGINEERING
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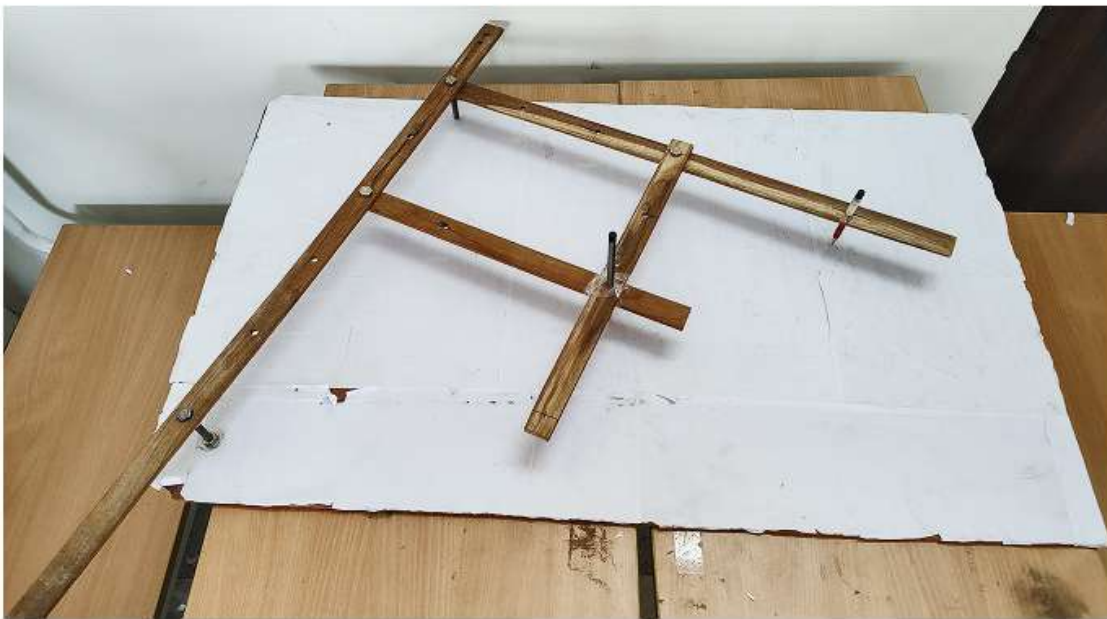
Pantograph

A pantograph is a planar four-bar linkage mechanism used for the geometric scaling (enlargement or reduction) of a given figure or motion.

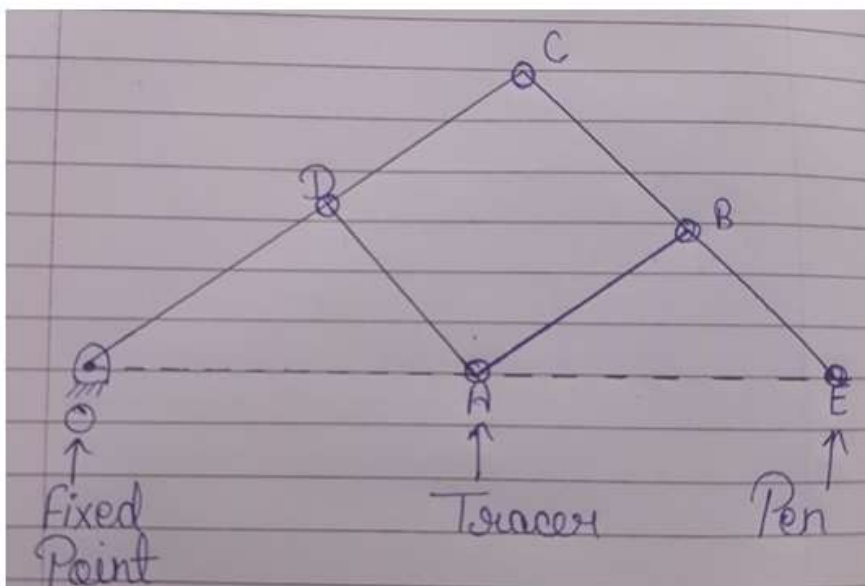
It consists of four rigid links connected by pivot joints such that opposite links remain equal and parallel ($AB = DC$ and $AD = BC$), forming a parallelogram configuration.

A pantograph is a simple four-bar linkage mechanism arranged in a parallelogram configuration.

A necessary geometric condition for the pantograph mechanism is that the fixed point (O), tracer point (A), and output point (E) must remain collinear to ensure accurate geometric scaling.



This is a basic reference diagram of working principles.

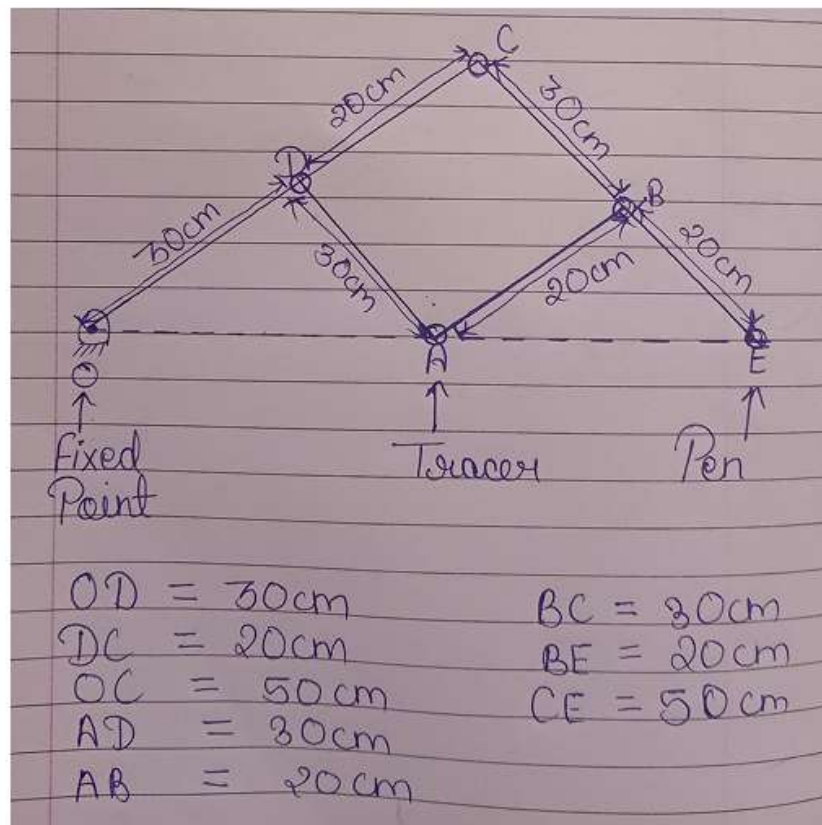


Assembly Rules

1. **Collinearity:** Ensure that the Fixed Point (O), the Tracer Point (A), and the Pen/Pencil Point (B) always lie on a single straight line.
2. **Parallelogram:** The interior links must form a perfect parallelogram ($AB = DC$ and $DA = CB$) for the scaling to remain accurate.

Scaling Ratio

- **Scale factor** = $\frac{\text{Distance to pen}}{\text{Distance to tracer}}$
 $= \frac{50}{30} = \frac{5}{3} = 1.67$ (approx)
- **Scale factor** = $\frac{\text{Distance to Pen}}{\text{Distance to tracer}}$
 $= \frac{60}{30} = \frac{6}{3} = 2$
 $\Rightarrow 1 : 2$



Calculations:

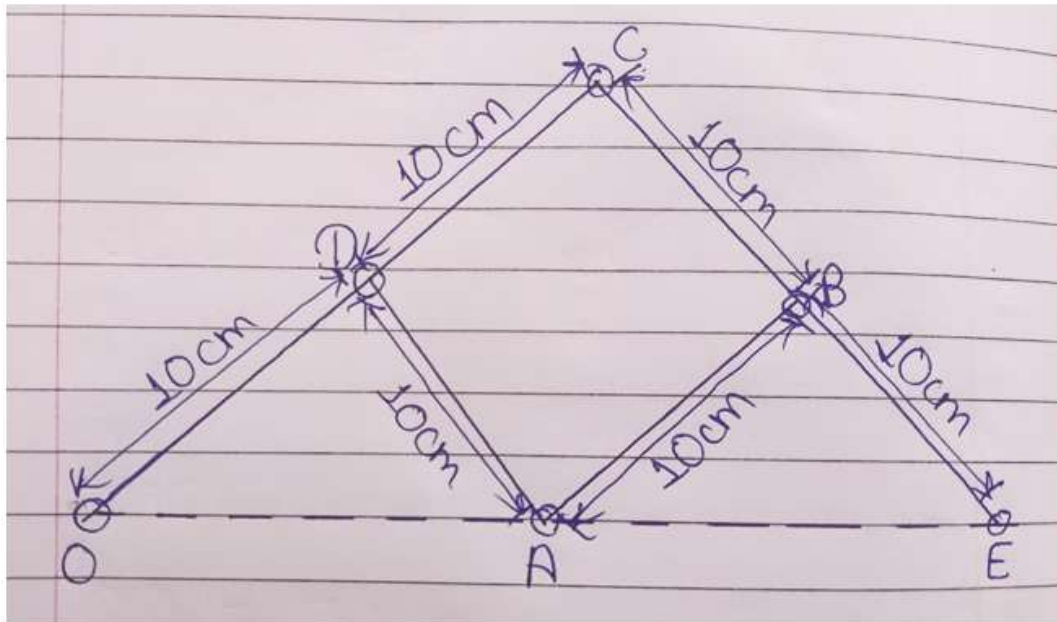
→ Calculation § 1:1.67 § 3

$$\frac{OD}{OC} = \frac{OA}{OE} = \frac{DA}{CE}$$

$$\frac{30}{50} = \frac{30}{50} = \frac{30}{50}$$

$$\frac{3}{5} = \frac{3}{5} = \frac{3}{5}$$

$$1.67 = 1.67 = 1.67$$



$$OD = 10 \text{ cm}$$

$$OC = 20 \text{ cm}$$

$$DC = 10 \text{ cm}$$

$$CE = 20 \text{ cm}$$

$$CB = 10 \text{ cm}$$

$$BE = 10 \text{ cm}$$

$$AD = 10 \text{ cm}$$

$$AB = 10 \text{ cm}$$

$$\frac{OD}{OC} = \frac{OA}{OE} = \frac{DA}{CE}$$

$$\frac{10}{20} = \frac{10}{20} = \frac{10}{20}$$

$$\frac{1}{2} = \frac{1}{2} = \frac{1}{2}$$

$$0.5 = 0.5 = 0.5$$

Using your specific formula:

$$\text{Output Length} = \text{Input Length} \times \text{Scale Factor}$$

Here is how your 6 cm trace would look for both ratios:

1. Example for 1:1.67 Ratio

In this configuration, your scale factor is 1.67. This is an enlargement.

- **Tracer Input:** 6 cm
- **Calculation:** $6 \text{ cm} \times 1.67$
- **Pen Output:** 10.02 cm

Note: As your notes suggest, this is an approximation ($5/3$). In a physical build, the output line would be exactly 10 cm if the links are perfectly aligned.

2. Example for 1:2 Ratio

In this configuration, your scale factor is 2. This doubles the size of the original image.

- **Tracer Input:** 6 cm
- **Calculation:** $6 \text{ cm} \times 2$
- **Pen Output:** 12 cm

